

Sessions 4-5 — Corporate Financial Risk Management: Managing Transaction Exposure

By the end of these sessions you must be able to name the main types of financial risk a firm faces, define the three types of foreign exchange exposure and explain why this material focuses on transaction exposure, work through the mechanics of the forward and options hedges for both a receivable and a payable with real numbers, compute a break-even spot rate between a forward and an options hedge, and explain what over-hedging is and why it creates losses of its own.

Plan

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1. Types of Financial Risk, and Why This Session Zooms Into One of Them

- A firm's overall "financial risk" is really a family of distinct risks, all interconnected: market risk, credit risk, liquidity risk, operational risk, foreign exchange (FX) risk, interest rate risk, business risk, reputational risk, and commodity risk.
- Understanding how these risks connect to each other matters as much as understanding each one individually — a shock in one (e.g. a commodity price spike) can cascade into others (e.g. liquidity risk, then reputational risk).

Example (A2A, an energy/utility group): the company tracks four financial risks side by side, each with its own causes and its own management levers.

- **Liquidity risk** — caused by tensions on capital markets, country risk, catastrophic events; managed with weekly financial planning, pre-funding activities, market diversification.
- **De-rating risk** (credit-rating downgrade risk) — caused by a worsening net financial position, worsening economic/financial ratios, or an imbalance between regulated and non-regulated business; managed with asset rotation, equity injections, hybrid bonds, dividend policy.
- **Interest rate risk** — caused by restrictive monetary policy or inflationary pressure; managed with hedging strategies and a balance between fixed- and variable-rate debt.
- **Sustainability covenant risk** — caused by failing sustainability performance targets or failing to allocate financial resources to eligible (green) investments; managed through a dedicated Sustainable Financial Committee.

- This session (and the next) zoom in on one specific risk from that list: **foreign exchange risk**, and more specifically the sub-type of FX risk called **transaction exposure** (defined in Section 3).

2. The Foreign Exchange Market: Spot, Forward and How Rates Are Quoted

- The FX market has (at least) two segments a firm needs to know before it can hedge anything:
- **Spot market** — almost immediate purchase or sale of foreign currency, at today's rate.
- **Forward market** — contracting **today** for a purchase or sale of foreign currency that actually happens on a **future** date. No money changes hands when the contract is signed; only at maturity.
- Forward contracts are used by companies and financial institutions either to **hedge** an existing FX exposure or to **speculate** on future exchange-rate moves.

Direct vs indirect quotation (from the point of view of a European investor): a **direct quotation** states how many units of home currency (€) buy one unit of foreign currency (\$) — e.g. €/\$. An **indirect quotation** states how many units of foreign currency (\$) buy one unit of home currency (€) — e.g. \$/€.

$$\text{Direct price} = 1 \div \text{Indirect price}$$

where:

- "Direct price" and "Indirect price" are simply the same exchange rate expressed from the two opposite directions.
- What this does: if you're given one quotation convention and the exam gives you the other, just take the reciprocal — don't re-derive the rate from scratch.

Forward premium / discount: the percentage difference between the spot rate and the forward rate for a currency, stated as an annualized percentage. If the forward rate for currency j is **above** the spot rate, currency j trades at a **forward premium** (the market expects it to strengthen); if **below**, it trades at a **forward discount**.

$$f_{N,j} = [F_N(\$ / j) - S(\$ / j)] \div S(\$ / j) \times (360 \div N)$$

where:

- $F_N(\$ / j)$ is the N-day forward rate for currency j, quoted as dollars per unit of j (a direct, "American terms" quote from the U.S. perspective).
- $S(\$ / j)$ is today's spot rate, same quotation convention.
- N is the number of days until the forward matures (e.g. 90 for a 3-month forward); 360 is the day-count convention used to annualize the result.
- What this actually does, step by step: (1) take the gap between the forward and spot rate; (2) divide by the spot rate to turn that gap into a percentage move; (3) multiply by $(360 \div N)$ to blow that percentage up to what it would be over a full year, so forward rates of different maturities become comparable on the same annualized scale. A positive result is a premium, a negative result is a discount.

Test Yourself, worked (Japanese yen vs U.S. dollar, 3-month forward): spot $S = \$0.00869/\text{¥}$, 3-month forward $F_{90} = \$0.00870/\text{¥}$, $N = 90$.
 $f = (0.00870 - 0.00869) \div 0.00869 \times (360 \div 90) = 0.0011507 \times 4 \approx 0.0046 = \mathbf{0.460\% \text{ forward premium}}$ — the yen is expected to strengthen against the dollar.

3. The Three Types of Foreign Exchange Exposure

Transaction exposure: the change in value of outstanding financial obligations that are already contractually fixed, caused by exchange-rate moves before those obligations are settled. It deals with actual cash flow changes on existing contracts.

Operating / economic exposure: the change in a firm's **expected future** cash flows (not yet contracted) caused by an exchange-rate move that affects the firm's international competitiveness — e.g. a stronger home currency makes its exports pricier abroad.

Translation / accounting exposure: the impact of FX volatility on the consolidation of foreign subsidiaries' accounts into the group's financial statements. It changes reported (consolidated) numbers but not actual cash flow.

- Transaction exposure — the focus of the rest of this session — typically arises from three kinds of business activity:
- Purchasing or selling goods/services priced and settled in a foreign currency.
- Borrowing or lending funds to be repaid in a foreign currency.
- Acquiring assets or incurring liabilities denominated in a foreign currency.

Hedging (transaction exposure): hedging reduces the variability of expected cash flows around their mean — that reduction in variance is what "reducing risk" means here. Hedging does not necessarily increase the expected cash flow itself; it narrows the range of possible outcomes.

4. Hedging Transaction Exposure: The Three Basic Choices

- Facing a transaction exposure, a firm has three basic alternatives: 1. **Remain unhedged** — do nothing, accept the FX rate that prevails on the settlement date. 2. **Hedge in the forward market** — lock in a rate today for the future transaction. 3. **Hedge in the options market** — pay a premium today for the **right** (not the obligation) to transact at a set rate.

4.1 The Forward Hedge

- If you will **owe** foreign currency in the future (you have a payable), enter a **long forward position**: agree today to **buy** the foreign currency at maturity, at the forward price. The buyer of a forward is obliged to buy the underlying at maturity at that pre-agreed price.
- If you will **receive** foreign currency in the future (you have a receivable), enter a **short forward position**: agree today to **sell** the foreign currency at maturity.

Worked example — hedging a payable with a long forward: a U.S. importer of British woollens owes £100 million in one year. Suppose the 1-year forward rate is \$1.50/£.

• **Unhedged:** if the pound depreciates to \$1.20/£, the importer buys £100m more cheaply and saves \$30m relative to \$1.50/£. If the pound appreciates to \$1.80/£, the importer pays \$30m more than at \$1.50/£. The unhedged outcome swings by ±\$30m depending on where the spot rate lands.

• **Hedged (long forward @ \$1.50/£):** the forward position's own gain/loss is the exact mirror image of the unhedged payable's gain/loss — if the pound falls to \$1.20, the forward loses \$30m (offsetting the unhedged saving); if the pound rises to \$1.80, the forward gains \$30m (offsetting the unhedged extra cost). **Combining the two, the hedged payable's total cost is flat at \$1.50/£ regardless of where the spot rate ends up** — that flat line is the entire point of hedging: it removes the uncertainty, not necessarily the average cost.

4.2 The Options Hedge

- Options give the same directional protection as a forward, but only use them when the market moves **against** you — you keep the upside if the market moves in your favor. This flexibility costs a premium, paid regardless of outcome.

- **Importers who owe foreign currency in the future should buy call options:** if the currency's price rises, the call locks in an upper limit on the dollar cost of the import; if the price falls, the importer simply lets the option lapse and buys the currency cheaper on the spot market.
- **Exporters with a foreign-currency receivable should buy put options:** if the currency's price falls, the put locks in a lower limit on the dollar value of the export; if the price rises, the exporter lets the option lapse and sells the currency at the better spot rate.

Watch out: the source also shows a generic "speculative forward position" payoff diagram (long vs short positions crossing at a reference forward price of \$1.0815/SFr, with illustrative gain/loss points near $\pm \$0.03/\text{SFr}$ at nearby spot rates). The diagram's own image made it ambiguous exactly which labeled point belongs to which line, so it isn't reproduced with attributed numbers here — the underlying mechanism is fully covered by the payable example above: long and short forward payoffs are always mirror images that cross zero exactly at the contracted forward rate.

5. Worked Example: Hedging a Foreign Currency Receivable — Aidan's Case

- Aidan's, a European company, sells equipment to Regency (a British firm) for £1,000,000. The sale happens in March; payment is due three months later, in June. This single sale is unusually large relative to Aidan's normal business, and Aidan has no other foreign customers — so this one receivable is a real concern.
- Market data available to Aidan's treasurer at the time of the sale:
- Spot rate: €1.09/£ (so the £1,000,000 sale is worth €1,090,000 at today's rate).
- 90-day forward rate: €1.0760/£.
- An FX advisor's 90-day forecast: €1.085/£.
- A June (3-month) put option to sell £1,000,000, strike €1.07/£, premium 1.5%.

The three hedge choices, worked through:

- **Remain unhedged:** do nothing in March; in June, sell £1,000,000 spot at whatever that day's rate turns out to be — euro proceeds are entirely uncertain today.
- **Forward hedge:** sell £1,000,000 forward today at €1.0760/£. In June, deliver the £1,000,000 against the forward sale and receive a **fixed €1,076,000**, whatever the spot rate does.
- **Options hedge:** buy the put to sell £1,000,000 at €1.07/£. Premium = $1.5\% \times €1,090,000$ (the sale's euro value at today's spot rate) = **€16,350**, paid upfront. In June, Aidan's either delivers the £1,000,000 against the put (receiving €1,070,000) or, if the spot rate has moved above €1.085/£, lets the put lapse and sells £1,000,000 on the spot market instead for a better euro amount.

- How to choose between the three: the final choice combines the firm's risk tolerance, its own view on where the pound is heading, and its confidence in that view. Managing transaction exposure with contractual hedges (forward, options) is ultimately a matter of managerial judgment, not a single "correct" formula-driven answer.

Watch out: the slide also lists Aidan's weighted average cost of capital (12% p.a.) and separate EUR/GBP 3-month borrowing and investment rates — these are the inputs needed for a fourth alternative, a **money market hedge** (borrow £ now against the future receivable, convert to € today, invest the € until June). The extracted slides only walk through the unhedged/forward/options comparison explicitly, so the money-market hedge numbers aren't worked through here — confirm with the instructor whether it was covered live as a fourth alternative.

6. Boeing Exercise: Forward vs Options, for a Receivable and for a Payable

6.1 Hedging a Receivable: The Landing-Gear Sale

- Boeing sells a landing-gear system to British Airways for £10 million, payable in one year. If the pound weakens before then, Boeing's dollar revenue shrinks.
- Data: forward rate \$1.46/£; a put option with strike \$1.46/£ costs a premium of \$0.02/£.
- Forward hedge:** guaranteed proceeds = $\$1.46 \times £10\text{m}$ = a flat **\$14.6 million**, regardless of the spot rate at maturity.
- Options hedge (put):** if the spot rate stays below the \$1.46 strike, exercise the put and net (strike – premium) $\times £10\text{m}$ = $(1.46 - 0.02) \times 10\text{m}$ = **\$14.4 million**. If the spot rate rises above the strike, let the put lapse and sell at the market rate instead, netting (spot – premium) $\times £10\text{m}$ — rising without limit as the pound strengthens.

Break-even spot rate (receivable, put hedge): $S^*_T = F + \text{premium}$

where:

- F is the forward rate, premium is the per-unit option premium, both quoted in the same \$/£ terms.
- What this does: above the break-even spot rate, the un-exercised put's payoff (spot minus premium) overtakes the forward's fixed payoff; below it, the forward wins. Setting the two payoffs equal and solving for the spot rate is exactly how the break-even is found: $\$1.46 + \$0.02 = \textbf{\$1.48/£}$, matching the source's own graph.

6.2 Hedging a Payable: The Rolls-Royce Engine

- Boeing now **imports** a Rolls-Royce jet engine for £5 million, payable in one year — the opposite exposure, so the correct option is a **call**, not a put (per the importer/exporter rule in Section 4.2).
- Data given: call option premium \$0.018/£, strike \$1.80/£. The source's own solved chart shows: forward hedge flat at **\$8,750,000** total cost (implying a forward rate of about \$1.75/£), the option hedge capped at **\$9,095,400** once the call is exercised, and a break-even spot rate of $S^*_T = \textbf{\$1.731/£}$.

Break-even spot rate (payable, call hedge): $S^*_T = F - \text{premium}$

where:

- Symmetric to the receivable case, but subtracted instead of added, because here the un-exercised call's cost (spot plus premium) needs to **fall below** the forward's fixed cost to become the cheaper choice, which happens at a **lower** spot rate than the forward.
- Checking it against the chart's own numbers: $\$1.75$ (implied forward rate) – $\$0.019 \approx \1.731 — consistent with the source's stated break-even, to a very close rounding.

Watch out: the slide's premium is printed as \$0.018/£, but working backward from the chart's own break-even (\$1.731) and cap (\$9,095,400) figures implies a premium closer to \$0.019/£ (and the missing money-market interest-rate table for this exercise, lost in PDF extraction, may explain the small remaining gap) — use the chart's final figures (\$1.731 break-even, \$9,095,400 cap, \$8,750,000 forward) as the authoritative numbers if this exact exercise reappears, not a from-scratch recomputation with the printed \$0.018.

7. Over-Hedging: When Hedging More Than You Actually Have Backfires

Over-hedging: hedging a larger foreign-currency amount than what actually materializes. It turns a hedge (meant to reduce risk) into a speculative position on the un-realized portion, because the firm is now contractually obligated to deliver currency it doesn't have from the underlying business and must go buy it in the market.

Worked example, step by step (a European company exporting to the US):

T0 (Budget): expected revenue = \$100 (100% in USD); budget rate = \$/€ 1.3 indirect (€/ \$ 0.77 direct);
expected costs = €70 (100% in EUR).
Expected margin = $(100\$ \times 0.77 \text{ €/\$}) - 70\text{€} = 77\text{€} - 70\text{€} = \text{€}7$.

Scenario check — how FX volatility alone would move that margin (holding the \$100 revenue assumption fixed):

- Strong € / weak \$ at \$/€ 1.6 (€/ \$ 0.625): margin = $(100\$ \times 0.625) - 70\text{€} = 62.5\text{€} - 70\text{€} = -7.5\text{€} \rightarrow$ unexpected loss of $-(7+7.5) = \text{-14.5}$.
- Strong \$ / weak € at \$/€ 1.0: margin = $(100\$ \times 1.0) - 70\text{€} = 100\text{€} - 70\text{€} = 30\text{€} \rightarrow$ unexpected gain of $30-7 = \text{+23}$.

Now the actual over-hedging case: at T0 the CFO sells 100% of the expected \$100 forward at 1.3 \$/€. But at T1 (Invoicing), actual revenue turns out to be only \$50, not \$100 — the company invoiced less than expected. The CFO is now **over-hedged**: contractually obligated to deliver \$100 forward but only actually earning \$50.

At T2 (Collection), suppose the spot rate is 1.0 \$/€ (a strong dollar):

- CFO receives \$50 from the real sale and converts it at the forward rate: $50\$ \times (1/1.3) = 38.5\text{€}$.
- But the forward contract obligates delivery of the full \$100, so the CFO must go buy the missing \$50 on the spot market at 1.0 \$/€ (i.e. pay €50 to get that \$50) and hand it over at the forward rate of 1.3 (worth only €38.5 that way) — a loss of $€50 - €38.5 = \text{-€11.5}$ purely from over-hedging.

Current margin = revenue – costs (including the over-hedging loss) = $38.5\text{€} - 11.5\text{€} - 70\text{€} = \text{-€43}$.

Total unexpected loss = $-(7 + 43) = \text{-€50}$, driven by over-hedging on top of the ordinary budgeted margin.

- The lesson: the loss above comes purely from the **speculative** position created by hedging currency that was never actually earned — it is not a normal FX-volatility loss like the -14.5/+23 examples above.
- This leaves two design questions every hedging policy must answer: 1. **When should hedging start** — at budget stage or only once the sale is actually invoiced (i.e. once the exposure is certain)? 2. **How much should be hedged** — 100% of the expected/budgeted amount, or some smaller, more conservative share?

8. Full Recap: Budget, Invoicing and Collection Together

- The whole hedging decision unfolds over three dates for a European company exporting to the US: **T0 (Budget), T1 (Invoicing), T2 (Collection)**.
- Data for the recap exercise: expected exposure (EE) at T0 = \$100; invoicing exposure (IE) at T1 = \$100 (confirmed, unchanged from budget in this version of the exercise); T0 spot rate = 1.5758 \$/€ (indirect) / 0.6346 €/ \$ (direct); T1 spot rate = 1.7 \$/€ / 0.5882 €/ \$; T0 forward = 1.54 \$/€ / 0.6494 €/ \$; T1 forward = 1.65 \$/€ / 0.6061 €/ \$; budget rate = 1.6 \$/€ / 0.625 €/ \$; T0 put option price = \$0.02 per \$ sold. At T2, the spot rate can land at either 1.4 or 1.75 \$/€, each with 50% probability.
- **At T0**, three alternatives are available: (1) do nothing; (2) sell \$100 forward at 1.54 \$/€ → locks in about €65; (3) buy a put with strike = T0 spot (1.5758 \$/€) and premium \$0.02 per \$ — option premium in euros = $0.02 \times 100\$ \times 0.6346 \text{ €/\$} = \text{€1.269}$.
- Companies genuinely choose "do nothing" (Alternative 1) when: competitors aren't hedging either, the FX risk is core to the business anyway, the firm has spare liquidity to absorb a loss, it can raise debt or equity if

needed, or the exposure can be diversified away by shareholders in their own portfolios.

- **At T1**, if the firm did nothing at T0, it faces the same choice again: keep waiting until T2, or sell \$100 forward at the **current** T1 forward rate (1.65). Selling at 1.65 is not automatically the best move — it "destroys" the value of the foreign revenue (and therefore the expected margin) relative to what was originally budgeted, since 1.65 is a worse rate for a euro-receiving exporter than the original budget rate of 1.6.
- **The put option floor, computed two ways** depending on which final scenario materializes (because the option is only exercised when it's the better outcome):
 - If T2 spot = 1.75 (a bad, strong-euro outcome for the exporter): exercise the put at the original strike, using the **T0** spot rate in the payoff: $\text{Floor} = [100\$ \times 0.6346 - \text{€}1.269] \div 100\$ = 0.6219 \text{ €/€} (1.60795 \text{ \$ /€}) \rightarrow \approx \text{€}62$.
 - If T2 spot = 1.4 (a good, strong-dollar outcome): don't exercise, sell on the market instead, so the payoff uses the **T2** spot rate: $\text{Floor} = [100\$ \times 0.7143 - \text{€}1.269] \div 100\$ = 0.715 \text{ €/€} (1.4253 \text{ \$ /€}) \rightarrow \approx \text{€}70$.

Final comparison (realized euro revenue), the whole point of the exercise:

- If T2 spot = 1.4 (good, strong \$): do nothing → €71.42 (**best case overall**); T0 forward → €64.93; T1 forward → €60.60; option → €70.15.
 - If T2 spot = 1.75 (bad, strong €): do nothing → €57.18 (**worst case overall**); T0 forward → €64.93; T1 forward → €60.60; option → €62.19.
 - Every one of these outcomes should be compared against the originally **budgeted** margin of €62.5, not just against each other.
- Reading the table: doing nothing has the widest range of outcomes (best in one scenario, worst in the other) — exactly the risk that hedging is meant to remove. The forward locked in at T0 (€64.93 either way) is the most stable, but it's inferior to "do nothing" in the good scenario and superior to it in the bad one. The option sits in between: it protects the downside (€62.19, close to the forward) while still capturing most of the upside (€70.15, close to "do nothing") — the premium is the price paid for keeping that upside.
 - Note on the direction of the quotation: T2 spot = 1.4 \$/€ is **lower** than 1.75 \$/€, meaning fewer dollars are needed to buy one euro, i.e. the dollar is **stronger**. Since this exporter earns dollars and needs to convert them into euros, a stronger dollar converts into **more** euros — hence 1.4 is the "good" scenario and 1.75 the "bad" one for this company, even though 1.75 is numerically the larger number.

9. Common Mistakes & Exam Traps

- Confusing transaction exposure (already-contracted cash flows) with operating/economic exposure (future, not-yet-contracted cash flows) or translation exposure (a consolidation/accounting effect with no direct cash-flow impact) — know which one a scenario is describing before picking a hedge.
- Getting the importer/exporter options rule backwards: importers (who **owe** FX) buy **calls**; exporters (who will **receive** FX) buy **puts**. Mixing this up gives exactly the wrong protection.
- Forgetting that an option's premium is paid regardless of outcome — it must be netted against the payoff either way, not treated as a cost that only applies when the option is exercised.
- Treating a forward hedge as something that improves the **expected** outcome — it doesn't; it removes variance around the mean, which can mean giving up upside as the price of removing downside.
- Sizing a hedge to the **budgeted** or **expected** exposure and then not adjusting it once the real (invoiced) exposure is known — this is exactly how over-hedging happens, turning a risk-reducing tool into a speculative position on the unrealized gap.
- Reading an indirect \$/currency quotation without checking which direction is "good news" for the company in question — whether a higher number is favorable depends on which currency the company earns versus which it reports in, not on whether the number itself went up or down.
- Comparing hedge outcomes only against each other and forgetting to also compare them against the originally budgeted amount, which is the actual benchmark the treasury team is trying to protect.

10. Must Know Absolutely

Must know:

- Direct price = $1 \div$ Indirect price.
- Forward premium/discount: $f_{N,j} = [F_N(\$ / j) - S(\$ / j)] \div S(\$ / j) \times (360 \div N)$; positive = premium (currency expected to strengthen), negative = discount.
- Three FX exposures: transaction (existing contracts), operating/economic (future competitiveness), translation (consolidation accounting only).
- Three hedge choices: remain unhedged, forward (locks a rate, no premium, symmetric gain/loss), options (pay a premium, keep the favorable-side upside).
- Importers (owe FX) buy **calls**; exporters (receive FX) buy **puts**.
- Break-even spot rate: receivable/put hedge $\rightarrow S_T^* = F + \text{premium}$; payable/call hedge $\rightarrow S_T^* = F - \text{premium}$.
- Over-hedging = hedging more than the currency amount actually realized; the un-covered portion becomes a speculative position, generating its own loss (or gain) on top of ordinary FX volatility.
- Two hedging-policy design questions: when to start hedging (budget vs invoicing), and how much to hedge (100% vs a smaller share).

Watch Out

Watch out: the "speculative forward position" payoff diagram (long vs short lines crossing at \$1.0815/SFr) could not be attributed to specific labeled values with full confidence from the extracted image — only the general mirror-image mechanism is used in this fiche, illustrated instead with the fully unambiguous \$100m/£/woolens importer example.

Watch out: Aidan's case lists WACC and separate EUR/GBP borrowing and investment rates that go unused in the unhedged/forward/options comparison extracted here — they look like the inputs for a fourth "money market hedge" alternative that may have been worked through live in class; confirm with the instructor and add the computation if so.

Watch out: in the Boeing payable exercise, the printed call premium (\$0.018/£) doesn't fully reconcile with the chart's own break-even (\$1.731) and cost-cap (\$9,095,400) figures, which imply a premium closer to \$0.019/£ — likely because the money-market interest-rate table referenced on that slide was lost in PDF extraction. Use the chart's final numbers as authoritative, not a recomputation from the printed premium.

Watch out: the Test Yourself questions on forward premium/discount (Section 2) and the four review questions in the Recap Exercise (transaction exposure definition, what FX derivatives hedge, which hedge locks in a dollar payable outflow, and the main risk of over-hedging) carry no marked correct answer in the extracted slides — the forward-premium example above was solved independently by applying the taught formula; the four review questions are conceptual restatements of Sections 3, 4 and 7 above and can be answered directly from those sections.